

Circuit Protection: Too Important to be an Afterthought



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Determining whether a circuit is adequately protected can require a high-level view of the electrical distribution system, from the fault current available at the source of supply down to the end device connected to the system. Circuit protection devices fulfil two primary purposes, namely, safety and reliability. Safety is assured by disconnecting the power in a circuit during over current, which eliminates electrocution and fire hazards.

In addition, the right protection may be necessary to comply with agency stand-

ards for some end products. Ranging from the classic fuse to schemes for protection against electrostatic discharge (ESD), these solutions are proof that simplicity still produces the best results. Thanks, freakonomics!

You take care of things valuable to you, monetary or otherwise. You may take care of your first love letter the same way a *Trekkie* would take care of his mint-condition Spock action figure. Closer home, the same level of diligent care is required to protect a small sub-circuit as in protecting a million-dollar sub-station. That is why circuit protection is immensely important (probably the biggest understatement in this article). Companies spend a lot in making sure their investments are safe. After all, what good is a device that does not work long-term or has the temperament of the South African cricket team in World Cup knockout games.

